Mufakkir: RAG-based Arabic Educational Chatbot for University Students

Mohamed Khaled Tawfik
Scientific Computing
Faculty of Computer and Information Science, Ain Shams University
Cairo, Egypt
2021170463@cis.asu.edu.eg

Mohamed Osama Ali
Scientific Computing
Faculty of Computer and Information Science, Ain Shams University
Cairo, Egypt
2021170454@cis.asu.edu.eg

Sohier Khaled Mohamed
Scientific Computing
Faculty of Computer and Information Science, Ain Shams University
Cairo, Egypt
2021170245@cis.asu.edu.eg

Farah Safwat Ezz Elregal
Scientific Computing
Faculty of Computer and Information Science, Ain Shams University
Cairo, Egypt
2021170392@cis.asu.edu.eg

Paula Emad Edward
Scientific Computing
Faculty of Computer and Information Science, Ain Shams University
Cairo, Egypt
2021170125@cis.asu.edu.eg

Maryam Elsawaf
Scientific Computing
Faculty of Computer and Information Science, Ain Shams University
Cairo, Egypt
maryam.sawaf@cis.asu.edu.eg

Dina Khattab
Scientific Computing
Faculty of Computer and Information Science, Ain Shams University
Cairo, Egypt
dina.khattab@cis.asu.edu.eg

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***Abstract* — The scarcity of Arabic-language educational tools tailored for university students, particularly in specialized fields like law and humanities, limits academic engagement and learning effectiveness. Mufakkir addresses this gap by providing an Arabic educational RAG-based chatbot designed to assist students with retrieving, summarizing, and interacting with academic content. The system is trained on a curated Arabic dataset of legal books collected from Ain Shams University, with synthetic question–context pairs generated for evaluation using an LLM. Mufakkir achieved 88% accuracy, 88% recall, and 0.78 nDCG@10 on this benchmark. Its embedding model ranked first on the Arabic RAG Leaderboard with a score of 0.65, reflecting strong retrieval performance. In answer generation, it outperformed major LLMs such as DeepSeek, LLAMA, and Qwen, scoring 0.79 in Answer-Ground Similarity and 0.73 in Question-Answer Similarity. By combining fine-tuned embeddings, hybrid retrieval, and domain-specific evaluation data, Mufakkir delivers accurate, context-aware educational support.**

Keywords*—*Educational Assistant, Retrieval-Augmented Generation (RAG), embedding model, question answering, Natural Language Processing (NLP).

I. Introduction

There is a growing need for Arabic-language educational systems capable of providing contextually relevant academic assistance. Existing tools [1],[2] often lack support for Arabic or fail to reflect the structure and terminology of localized curricula, particularly in fields like law and the humanities, resulting in reduced engagement and learning effectiveness. This paper introduces Mufakkir, an Arabic-language assistant built using Retrieval-Augmented Generation (RAG) and Natural Language Processing (NLP) techniques, trained on curated academic content. Mufakkir is designed to be accurate, context-aware, and accessible for students in Arabic-speaking academic environments, offering features such as question answering, document summarization, file upload with content extraction and answering, and quiz generation.

Mufakkir represents a significant advancement in Arabic NLP by delivering a tailored educational chatbot that addresses the critical shortage of context-aware, curriculum-specific tools for Arabic-speaking university students. The main contributions of this work include the introduction of a fine-tuned embedding model optimized for Arabic academic texts, along with a hybrid retrieval strategy combining semantic and lexical approaches. The system provides robust features such as question answering, document summarization, and quiz generation to enhance student engagement and learning effectiveness in fields such as law and humanities. To the best of our knowledge, this is the first work that presents an Arabic-enabled RAG-based model designed specifically for educational assistance..

Another key innovation of this work lies in the construction of a domain-specific dataset; MufakkirTriples which is built from Arabic law books taught for university students. To enable high-quality evaluation, we generated synthetic question–context pairs using a large language model, producing a benchmark tailored to the law domain to fill a critical gap in Arabic-language academic NLP resources.

In terms of performance, Mufakkir ranked first on the Arabic RAG Leaderboard outperforming existing models such as DeepSeek [3], LLaMA [4], and Qwen [5]. In answer generation, it also achieved higher scores in Answer-Ground Similarity and Question-Answer Similarity, demonstrating its superiority in producing accurate and relevant responses. These results confirm Mufakkir's effectiveness in delivering contextually grounded, educationally aligned answers within low-resource domains like Arabic law.

The paper is structured as follows: Section II reviews related work on Arabic language assistants, RAG systems, and educational NLP applications to position Mufakkir within the current research. Section III explains the methodology, covering query processing, document chunking, embedding model fine-tuning, hybrid retrieval, and techniques to reduce hallucination. Section IV outlines the dataset, including its collection, preprocessing, and the creation of 25,000 question-answer-context triples. Section V highlights the evaluation of Mufakkir's performance in comparison to existing models. Finally, Section VI concludes with a discussion of the system's contributions, current limitations, and future research opportunities.

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II. Related Work

A. Arabic Language Assistants

The development of intelligent assistants for Arabic-speaking users has gained traction due to the need for contextually relevant tools. Existing systems often leverage NLP to process Arabic text, focusing on tasks like question answering and text understanding. However, many lack support for localized educational content, particularly in specialized fields such as law and humanities, leading to reduced effectiveness in academic settings.

Sharma et al. [1] presented aiXplain SDK, a toolkit designed to standardize the development of AI assets, including Arabic NLP systems for tasks like question answering. The toolkit supported multilingual processing but focused primarily on general-purpose applications, lacking deep integration with curriculum-specific academic content, such as legal or humanities texts, which limits its utility for university students. Almazrouei et al. [2] introduced the Falcon series of open language models, which demonstrated strong performance in Arabic text processing. However, these models were not optimized for educational contexts, and their general-purpose nature resulted in limited handling of domain-specific terminology, reducing their effectiveness for specialized academic tasks.

B. Retrieval-Augmented Generation for Arabic

Retrieval-Augmented Generation (RAG) emerged as a powerful approach for enhancing language models by integrating external knowledge retrieval, improving factual accuracy and context awareness. In the context of Arabic, RAG systems combined semantic embeddings and generative models to handle linguistic complexities like morphology and dialect variations. These systems are versatile but often not tailored for educational applications. The evolution of RAG built on advancements in transformer-based Large Language Models (LLMs), such as those introduced by Vaswani et al. [6]. They proposed the transformer architecture, enabling efficient natural language understanding for multilingual tasks, including Arabic. However, their work focused on model architecture rather than retrieval integration, limiting its direct applicability to RAG systems.

El-Beltagy [7] explores RAG for Arabic, focusing on its ability to handle linguistic nuances through semantic embeddings. The study demonstrated improved retrieval for general Arabic texts but lacked optimization for academic domains, resulting in limited relevance for educational applications. Alsubhi [8] conducts a systematic analysis of RAG pipeline components for Arabic, identifying optimal configurations for retrieval and generation. While comprehensive, the study did not address educational use cases, and its generic approach missed curriculum-specific requirements.

C. Semantic Search in Arabic RAG

Semantic search in Arabic RAG systems leveraged dense vector representations to retrieve relevant documents by capturing semantic relationships between queries and texts. These systems used embeddings to encode queries and documents, enabling similarity-based retrieval that handled linguistic nuances like synonyms and morphological variations in Arabic. Techniques such as cosine similarity and hybrid retrieval, combining semantic and keyword-based approaches, enhance retrieval accuracy for complex academic content. The study in [9] evaluated semantic search in Arabic RAG, highlighting its role in improving context-aware retrieval. It demonstrated high retrieval accuracy for general Arabic texts but notes challenges in handling domain-specific academic content due to limited dataset diversity. Alroumi [10] further explored optimized embedding training for Arabic semantic search, achieving improved retrieval performance. However, the approach is computationally intensive and not specifically tailored for educational applications.

D. Fine-Tuning Embeddings for Arabic

Fine-tuning embeddings is a key strategy for improving text understanding in Arabic RAG systems. By optimizing sentence representations, these methods enhanced the accuracy of retrieval and generation tasks, particularly for educational tools processing academic content. Alroumi [10] investigated embedding fine-tuning for Arabic RAG, proposing techniques to optimize semantic similarity in high-dimensional spaces. While effective, the method required significant computational resources and lacked focus on educational contexts, limiting its direct applicability to student-facing tools. Nacar et al. [11] explored nested embedding learning to enhance semantic similarity in Arabic NLP, achieving robust performance in text understanding. However, their approach is not tailored for curriculum-specific tasks, and its complexity may hinder scalability in resource-constrained settings.

E. Educational Applications of Arabic NLP

NLP-based educational tools for Arabic focus on tasks like language learning and content analysis. While some systems support features like poetry analysis or dialect identification, few address curriculum-specific needs in academic settings. Sharma et al. [1] described aiXplain SDK's applications in educational NLP, including Arabic language learning tools. However, its general-purpose design lacked deep integration with academic curricula, limiting its effectiveness for specialized fields like law. Nacar et al. [11] propose NLP methods for educational content analysis, demonstrating potential in text summarization. Their work, however, focused on broad applications and did not address the specific needs of university students in disciplines like humanities.

III. METHODOLOGY

Mufakkir's system architecture integrates multiple components to deliver a robust Arabic educational chatbot tailored for university students. The system combines query preprocessing, document chunking, a fine-tuned embedding model for hybrid retrieval, a cross-encoder re-ranker, and an LLM for answer generation, with mechanisms to prevent hallucinations. User queries are preprocessed for accurate interpretation, followed by retrieval of relevant academic content from a vector database of chunked documents. The fine-tuned Mufakkir_Embedding model, optimized for Arabic academic texts, enables hybrid retrieval combining semantic and lexical approaches. Retrieved documents are reranked for relevance, and the Qwen2-72B LLM [5] generates context-aware responses, summaries, or quizzes, validated against retrieved context to ensure factual accuracy. This architecture, illustrated in Fig. 1, ensures efficient and accurate support for educational tasks. The following subsections detail each component of the system.

A. Query Processing

The system preprocesses user queries to ensure accurate interpretation and effective retrieval of relevant academic content. This involves a multi-step pipeline designed to handle the linguistic complexities of Arabic text, including punctuation normalization, stop-word removal, tokenization, and query expansion. Punctuation normalization removes or standardizes punctuation marks (e.g., converting multiple consecutive punctuation marks to a single mark) to reduce noise and ensure consistent query formatting, which is critical for matching Arabic queries with varied typographical styles. Stop-word removal eliminates common Arabic words that carry minimal semantic value, using a curated stop-word list tailored for Arabic academic texts to enhance retrieval focus on meaningful terms. Tokenization splits the query into individual tokens, leveraging an Arabic-specific tokenizer to handle morphological variations, such as verb conjugations and noun inflections, ensuring accurate segmentation of complex words. Query expansion is occasionally applied to include synonyms or related terms using a domain-specific thesaurus derived from the academic dataset, improving recall for

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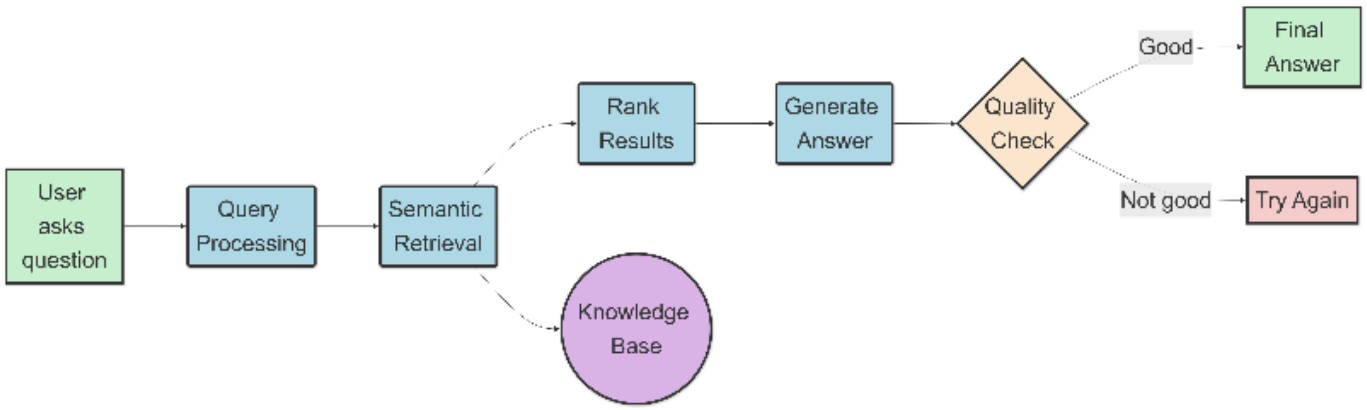


Fig. 1. System Architecture Layers

semantically similar documents. These steps collectively ensure precise user intent analysis, enabling robust retrieval and generation for Arabic educational queries, as described in [7].

B. Document Chunking

The system processes a corpus of over 80 Arabic academic books, primarily legal and humanities texts, by segmenting them into smaller, coherent chunks to facilitate efficient and accurate retrieval. This is achieved through a recursive chunking method, which hierarchically splits documents to maintain semantic integrity and achieve an optimal chunk size. Initially, each book is divided into sections based on structural markers, such as chapters or headings, identified using metadata extracted via Azure Document Intelligence. Each section is then recursively subdivided into smaller units, typically paragraphs or sentence groups, ensuring that each chunk contains semantically cohesive content, such as a complete legal argument or a thematic discussion in humanities. The chunk size is optimized to balance granularity and context, targeting approximately 200–300 words per chunk to align with the embedding model's input constraints and ensure compatibility with the vector database. To preserve semantic integrity, the algorithm avoids splitting sentences or key phrases, using natural language processing techniques to detect sentence boundaries and semantic units in Arabic text, which is particularly challenging due to its morphological complexity. These chunks are then indexed in a Chroma DB vector database, where each chunk is encoded as a dense vector using the Mufakkir_Embedding model. Metadata, such as book title, chapter, and page number, is attached to each chunk to enable traceability and enhance retrieval precision. This approach ensures that the system can efficiently retrieve contextually relevant content for user queries, as described in Fig. 2.

C. Embedding Model Fine-Tuning

The core retrieval component is the Mufakkir_Embedding model, fine-tuned using the model of Triplet-Matryoshka-V2 [12] for academic content. The dataset consists of real and synthetic academic texts, leveraging two loss functions to optimize semantic similarity in a 768-dimensional embedding space. The fine-tuning process was conducted using a NVIDIA L4 GPU, providing efficient training throughput for high-dimensional embedding optimization. The Multiple Negatives Ranking Loss L maximizes the similarity between anchor and positive embeddings while treating other batch embeddings as negatives, defined as:

$$L = -\frac{1}{N}\sum_{i=1}^{N} \log\left(\frac{\exp(sim(e_i^a, e_i^p)\cdot \text{scale})}{\sum_{j=1}^{N}\exp(sim(e_i^a, e_j)\cdot \text{scale})}\right) \quad (1)$$

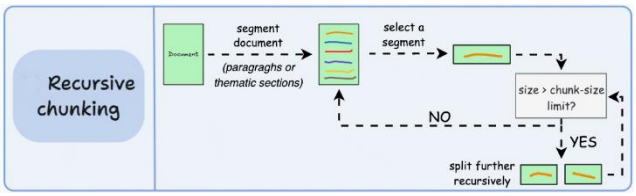


Fig. 2. Recursive Chunking

where N is the batch size, e_i^a and e_i^p are the anchor and positive embeddings for the i-th sample, e_j includes all batch embeddings, sim is the cosine similarity, and scale adjusts score sharpness. The workflow computes anchor-positive and anchor-negative similarities, applies log-SoftMax to maximize correct positive identification, and benefits from larger batch sizes for robust negative sampling [12].

The Matryoshka Loss *Lk* supports variable embedding dimensions by applying the base loss (Multiple Negatives Ranking Loss) across multiple reduced dimensions. For each dimension dk ∈ {768, 512, 256, 128, 64}, the loss is:

$$L_k = w_k \cdot L\left(e_{d_k}^a, e_{d_k}^p, e_{d_k}^n\right) \quad (2)$$

The total loss is *L(total):*

$$L_k = w_k \cdot L\left(e_{d_k}^a, e_{d_k}^p, e_{d_k}^n\right) \quad (3)$$

where K is the number of dimensions, ω_k is the weight for dimension dk, and $e_{dk}^a\, e_{dk}^p e_{dk}^n$ are the anchor, positive, and negative embeddings at dimension d_k. Embeddings are truncated at each dimension, and the base loss is computed and scaled. This enables flexible embedding sizes for retrieval tasks, enhancing performance across varying computational constraints [13].

D. Semantic Retrieval (Hybrid Approach)

A hybrid retrieval strategy, identified as the optimal approach, combines vector-based semantic retrieval with lexical keyword-based retrieval to maximize relevance. Fig. 3 illustrates the semantic retrieval pipeline, where the query and document chunks are passed through the fine-tuned Mufakkir_Embedding model for vectorization, followed by similarity search in the Chroma vector database. Semantic retrieval uses the fine tuned Mufakkir_Embedding model to encode queries and document chunks as dense vectors in Chroma DB, calculating similarity via cosine similarity, defined as:

$$\cos(q,d) = \frac{q.d}{||q||||d||} \quad (4)$$

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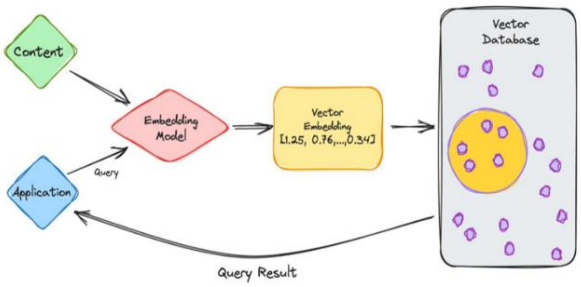


Fig. 3. Semantic Retrieval Pipeline

where q and d are the query and document vectors respectively. Lexical retrieval employs the BM25 algorithm, which scores documents based on term frequency and inverse document frequency:

$$\mathrm{BM25}(q,d) = \sum_{i=1}^{n} \mathrm{IDF}(q_i) \cdot \frac{f(q_i,d)\cdot(k_1+1)}{f(q_i,d)+k_1\cdot\left(1-b+b\cdot\frac{|d|}{\mathrm{avgdl}}\right)} \quad (5)$$

where q_i is a query term, $f(q_i, d)$ is the term frequency in document d, $IDF(q_i)$ is the inverse document frequency, $|d|$ is the document length, $avgdl$ is the average document length, and k1 and b are parameters. The Ensemble Retriever merges results with equal weights (0.5, 0.5), deduplicates, and sorts them to prioritize relevant matches [14].

E. Document Reranking

The system employs a cross-encoder re-ranker to enhance the relevance of retrieved documents by scoring query-document pairs and sorting them in descending order of relevance. Unlike bi-encoder models [13], which independently encode queries and documents into separate embeddings and compute similarity (e.g., via cosine similarity), the cross-encoder takes both the query and document as a single input, processing them jointly through a transformer-based architecture to capture deep contextual interactions. This joint encoding allows the cross-encoder to model complex semantic relationships, such as syntactic dependencies and contextual nuances in Arabic academic texts, resulting in more accurate relevance scores. For each query-document pair, the cross-encoder outputs a scalar relevance score, typically between 0 and 1, which is used to rank the top retrieved documents. The highest-scoring documents are passed to the generation phase, ensuring that only the most contextually relevant content informs the Qwen2-72B LLM's responses. The cross-encoder's superior accuracy over bi-encoder models stems from its ability to leverage full query-document interactions, though it is computationally more intensive. In Mufakkir, this trade-off is justified by the need for precise retrieval in academic settings, where relevance is critical for tasks like answering complex legal or humanities questions. This approach significantly improves retrieval performance, as described in [8].

F. Answer Generation

The Qwen2-72B LLM, a transformer-based large language model with 72 billion parameters, generates answers, summaries, or quizzes based on the user query and the top-ranked documents retrieved from the dataset. The model's architecture leverages advanced attention mechanisms to process input sequences, enabling it to understand and generate coherent text in multiple languages, including Arabic. In Mufakkir, the Qwen2-72B takes as input the preprocessed user query concatenated with the context from reranked document chunks, typically formatted as a prompt that instructs the model to produce a specific output type (e.g., a direct answer, a concise summary, or a set of quiz questions with answers). For instance, for a query requesting a summary of a legal concept, the model generates a condensed explanation by synthesizing relevant information from the retrieved chunks. Its multilingual capabilities are particularly suited for complex Arabic academic tasks due to its training on diverse datasets, including Arabic texts. This enables handling linguistic nuances such as morphological variations and domain-specific terminology in legal and humanities contexts.

The model's large parameter size enhances its ability to generate contextually accurate and fluent responses, critical for educational applications where precision and clarity are paramount. Additionally, the Qwen2-72B's fine-tuning for instruction-following ensures that it adheres to the prompt's requirements, producing outputs tailored to the user's intent, such as generating multiple-choice quizzes for student practice. This approach ensures robust performance for Mufakkir's educational features, as described in [8].

G. Hallucination Prevention

A specialized prompt template instructs the LLM to validate responses against retrieved context, regenerating or revising answers if inconsistencies are detected. This ensures factual integrity is critical for academic applications [5].

H. Summary

Mufakkir's methodology integrates query preprocessing, recursive document chunking, and a finetuned embedding model for hybrid retrieval. A Cross encoder re-ranker refines results, and the Qwen2-72B LLM generates context-aware outputs. Hallucination control ensures factual accuracy, making the system robust for Arabic educational tasks.

IV. DATASET

A. Collection Challenges

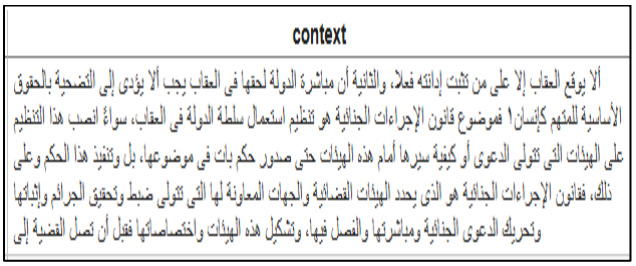
Creating a domain-specific Arabic academic dataset posed challenges due to the limited availability of structured educational resources. Manual collection of materials from faculties was time-consuming, particularly for non-digitized texts. The absence of standardized Arabic datasets for question answering and semantic search necessitated synthetic data generation to simulate academic scenarios [15].

B. Dataset Description

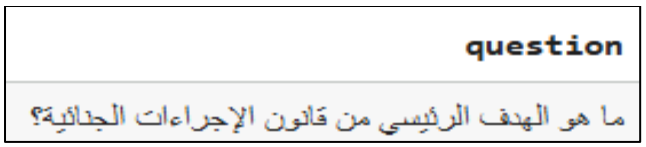
Our MufakkirTriples dataset comprises 80 academic books and lecture notes from various university departments, primarily focusing on legal and humanities texts, yielding approximately 25,000 question–answer–context triples. Each triple consists of a question, a corresponding answer, and a context derived from a document chunk, designed to reflect real-world challenges faced by Arabic-speaking university students. For example, a triple might include a question about the conditions for a valid contract in Egyptian law, an answer summarizing the legal requirements, and a context extracted from a relevant section of a legal textbook.

Fig. 4 presents an example data sample from the MufakkirTriples dataset, illustrating (a) a context excerpt from an academic textbook, (b) the generated question based on the context, and (c) the corresponding academic answer. These triples were generated using Qwen 70B Instruct with a carefully designed prompt template to ensure academic relevance, answer accuracy, and semantic coherence with the source document.

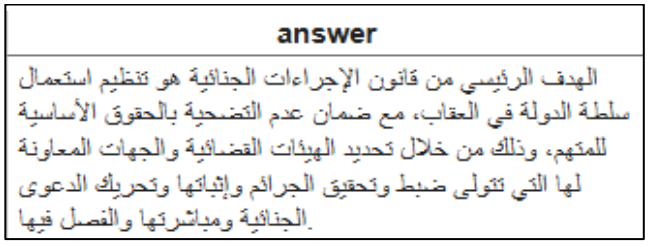
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a.



b.



c.

Fig. 4. Data sample from the MufakkirTriples dataset including a) excerpt from an academic textbook b) the query example generated from the context, and c) the corresponding academic answer.

The newly collected dataset supports fine-tuning the Mufakkir_Embedding model for improved semantic retrieval and enabled tasks like question answering and quiz generation, as described in [5]. The MufakkirTriples dataset is not publicly available due to copyright and privacy restrictions, as it includes excerpts from university materials. A limited, non-copyrighted subset may be shared for research purposes upon request.

C. Dataset Preprocessing

The MufakkirTriples dataset is prepared through a multi-step preprocessing pipeline to ensure quality and suitability for retrieval tasks. Documents are segmented into coherent chunks using recursive chunking to maintain semantic integrity and enable efficient retrieval, resulting in approximately 50,000 chunks across the 80 academic books and lecture notes. Each chunk is targeted to contain around 600 tokens, a size chosen to balance semantic coherence with the input constraints of the Mufakkir_Embedding model and the Chroma DB vector database. An overlap of 200 tokens is applied between consecutive chunks to ensure contextual continuity, preventing the loss of semantic connections across chunk boundaries. This overlap means that the last 200 tokens of one chunk are included as the first 200 tokens of the next chunk, allowing for seamless retrieval of related content when queries span chunk boundaries.

Azure Document Intelligence [16] is employed to extract structured content from PDFs and scanned materials, preserving academic layouts such as headings, sections, and tables, which are critical for maintaining the integrity of academic texts. Each chunk is annotated with metadata, including book titles, chapter identifiers, and page numbers, to enhance response traceability and trustworthiness, ensuring that retrieved content can be linked back to its source for verification [9].

V. EXPERIMENTS & RESULTS

A. Embedding Model Comparison

The Mufakkir_Embedding model is evaluated against State-Of-The-Art Arabic embedding models using information retrieval evaluation tools from Sentence Transformers [17] on a university-level academic dataset. Table I shows the comparative results for key metrics, including Accuracy@5, Recall@5, Normalized Discounted Cumulative Gain (nDCG@10), and Recall@5 quantifies the fraction of relevant items retrieved among all relevant items, which is defined as:

$$\text{Recall@5} = \frac{Number\ of\ Relevant\ Results\ Retrieved\ in\ Top\ 5}{Total\ Number\ of\ Relevant\ Items} \quad (6)$$

nDCG@10 evaluates ranking quality by considering the position and relevance of retrieved items, calculated as:

$$\text{BM25}(q,d) = \sum_{i=1}^{n} \text{IDF}(q_i) \cdot \frac{f(q_i,d)\cdot(k_1+1)}{f(q_i,d)+k_1\cdot\left(1-b+b\cdot\frac{|d|}{\text{avgdl}}\right)} \quad (7)$$

We report retrieval performance using metrics such as nDCG@10 and Recall@10. Here, "@10" refers to evaluating the model's effectiveness over the top 10 retrieved documents. nDCG@10 (Normalized Discounted Cumulative Gain at rank 10) captures both relevance and position, giving higher scores to relevant documents appearing earlier in the ranking. Recall@10 measures the fraction of all relevant documents that appear in the top 10 results. Higher values indicate stronger retrieval accuracy and ranking quality in limited output scenarios. Accuracy measures the proportion of correctly retrieved items among the top 5 results:

$$\text{Accuracy@5} = \frac{Number\ of\ Correctly\ Retrieved\ Items\ in\ Top\ 5}{Total\ Number\ of\ Queries} \quad (8)$$

Table I compares Mufakkir_Embedding with four leading Arabic embedding models: multilingual-e5-large-instruct [18], nomic-embed-text-v2-moe [19], Arabic-Retrieval-V1.0 [10], and Arabic-Triplet-Matryoshka-V2 [12].

According to Table I, Mufakkir_Embedding achieved the highest performance across all three metrics (Acc@5 = 0.88, Recall@5 = 0.88, nDCG@10 = 0.78). This demonstrates its ability to consistently retrieving the correct results within the top 5 and ranks them more effectively than the other models. The closest competitor, multilingual-e5-large-instruct [16], achieved solid performance (Acc@5 = 0.82, Recall@5 = 0.82, but lagged behind in ranking quality (nDCG@10 = 0.70).

TABLE I. COMPARISION OF ARABIC EMBEDDING MODEL PERFORMANCE

Model	Acc@5	Recall@5	nDCG@10
Mufakkir_Embedding [Ours]	0.88	0.88	0.78
multilingual-e5-large-instruct [18]	0.82	0.82	0.70
nomic-embed-text-v2-moe [19]	0.80	0.80	0.69
Arabic-Retrieval-V1.0 [10]	0.79	0.79	0.69
Arabic-Triplet-Matryoshka-V2 [12]	0.73	0.73	0.62

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The nomic-embed-text-v2-moe model [19] showed comparable performance to Mufakkir_Embedding in both Accuracy@5 and Recall@5 (0.80 vs. 0.88), but slightly underperformed in nDCG@10 (0.69 vs. 0.78), indicating weaker ranking of relevant results. Meanwhile, Arabic-Retrieval-V1.0 [10] and Arabic-Triplet-Matryoshka-V2 [12] had the lowest scores, suggesting they are less effective for academic retrieval tasks in this dataset.

Overall, Mufakkir_Embedding clearly outperformed all other models in both retrieval effectiveness and ranking precision, indicating its suitability for high-stakes academic applications in Arabic. The consistently higher nDCG@10 shows that not only does it find the right results, but it ranks them in a way that maximizes user satisfaction.

B. Leaderboard Evaluation

Mufakkir_Embedding, openly available on Hugging Face [20], has been submitted to the Arabic RAG Leaderboard, which evaluates over 45 embedding models' systems on a private Islamic dataset. The leaderboard provided a robust platform to test and validate the model's performance in retrieving and generating accurate Arabic academic content . Table II shows Mufakkir's first-place ranking with an Islamic Knowledge Score of 65.56, confirming its robustness and precision.

C. Answer Generation Evaluation

The evaluation of Mufakkir's answer quality is conducted using tools from the Sentence Transformers benchmark [15] on 400 challenging academic questions. The assessment utilized three metrics. Answer-Ground Similarity (AGS), which is an embedding-based comparison of generated answers with reference answers, calculated according to equation 9, where Agen is the embedding of the generated answer, and Aref is the embedding of the ground-truth reference answer, All three metrics are computed using the cosine similarity function Cos(x, y), which measures the angular similarity between two embedding vectors.

$$\mathrm{AGS} = \cos(A_{\mathrm{gen},} A_{\mathrm{ref}}) \tag{9}$$

Question-Answer Similarity (QAS), which is a measure of relevance between the generated answer and the input query, computed according to equation 10, where Q is the embedding of the input question, and Agen is the embedding of the generated answer.

$$\mathrm{QAS} = \cos(\mathrm{Q}, A_{\mathrm{gen}}) \tag{10}$$

And Faithfulness (F), which evaluates the alignment of generated answers with retrieved context to detect hallucinations, defined as equation 11, where C is the embedding of the retrieved context used to answer the question and Agen is the embedding of the generated answer. A higher faithfulness score indicates that the generated answer closely adheres to the supporting evidence, reducing the risk of hallucination.

$$F = \cos(A_{\mathrm{gen},} \mathrm{C}) \tag{11}$$

TABLE II. ARABIC RAG LEADERBOARD RESULTS

Model	Score
Mufakkir_Embedding [Ours]	0.65
Arabic-Retrieval-V1.0 [10]	0.64
Jina embeddings v3 [21]	0.62
AraModernBert-Base-STS [22]	0.62
Arabic-Triplet-Matryoshka-V2 [12]	0.59

As shown in Table III, Mufakkir outperformed DeepSeek [3], LLaMA [4], and Qwen [5] in Answer Generation Score (AGS) (0.79) and Question Alignment Score (QAS) (0.73). However, it scored lower in Faithfulness (F) (0.26), indicating the need for further improvements in ensuring answer reliability. These results demonstrate Mufakkir's strength in accuracy and relevance while emphasizing ongoing challenges in maintaining factual consistency.

Table IV presents the results of LLM-based human alignment judgment using the DeepSeek r1 model [3], where responses were scored on a 5-point Likert scale across three dimensions: coherence, completeness, and relevance, with 1 being poor and 5 being excellent. The evaluation aimed to simulate human judgment to assess how well each model delivers informative and well-structured answers.

Mufakkir demonstrated strong performance in higher-quality categories, achieving a peak of 168 responses rated as level 4, the highest among all models. Additionally, it secured 80 responses at level 5, reflecting a high degree of fluency and content alignment. In contrast, Qwen [5] and DeepSeek [3] had more responses clustered in the mid-range (level 3), while LLaMA [4] had the highest count in the lowest-quality band (58 responses at level 1), indicating lower output quality.

This distribution suggests that Mufakkir consistently generates well-grounded and relevant responses, with minimal hallucination or irrelevant output, particularly in academic domains. These results highlight Mufakkir's advantage in producing responses that score highly across all quality dimensions, confirming its effectiveness as an educational assistant.

TABLE III. ANSWER GENERATION PERFORMANCE

Metric	Mufakkir [Ours]	Qwen [5]	Llama [4]	Deep Seek [3]
Answer Ground Similarity	0.79	0.64	0.63	0.66
Question-Answer Similarity	0.73	0.70	0.68	0.71
Faithfulness Score	0.26	0.47	0.50	0.18

TABLE IV. LLM JUDGE SCORE

Metric	Mufakkir [Ours]	Qwen [5]	DeepSeek [3]	Llama [4]
1	9	44	42	58
2	28	118	82	168
3	115	135	88	118
4	168	82	97	43
5	80	21	91	13

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VI. CONCLUSION & FUTURE WORK

The introduction of Mufakkir represents a groundbreaking advancement in the field of Arabic NLP, specifically tailored to the educational domain. By integrating State-Of-The-Art RAG techniques, the chatbot has successfully combined fine-tuned embedding models, hybrid retrieval strategies, and advanced large language models to deliver superior performance in both retrieval and generation tasks. The system's notable achievements, including first place on the Arabic RAG Leaderboard and high scores in metrics such as Accuracy@5, Recall@5, and nDCG@10, underscore its technical robustness and practical relevance.

In answer generation, Mufakkir has shown exceptional capabilities in producing context-aware responses that align closely with user queries, as evidenced by its high scores in AGS and QAS. These strengths position Mufakkir as a pivotal tool for enhancing Arabic academic learning by bridging the gap in NLP tools for Arabic-speaking users. However, challenges remain, particularly in improving Faithfulness, where ensuring that generated responses adhere strictly to retrieved and factual information is a critical next step.

Looking ahead, future research will explore ways to address these limitations, including refining the system's faithfulness metrics and extending its applicability across a wider range of academic disciplines. Another important avenue of development is the incorporation of multilingual capabilities, enabling Mufakkir to serve diverse linguistic audiences and foster cross-cultural academic engagement. Additionally, expanding its dataset to encompass more specialized and interdisciplinary academic content will further strengthen its utility as an educational resource.

Ultimately, Mufakkir's contributions not only demonstrate the potential of RAG systems in advancing Arabic NLP but also lay the foundation for developing intelligent, accessible, and impactful educational tools for Arabic-speaking learners worldwide. Its success highlights the transformative power of integrating cutting-edge AI technologies into education, paving the way for future innovations that address the unique needs of underrepresented languages in the digital age.

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